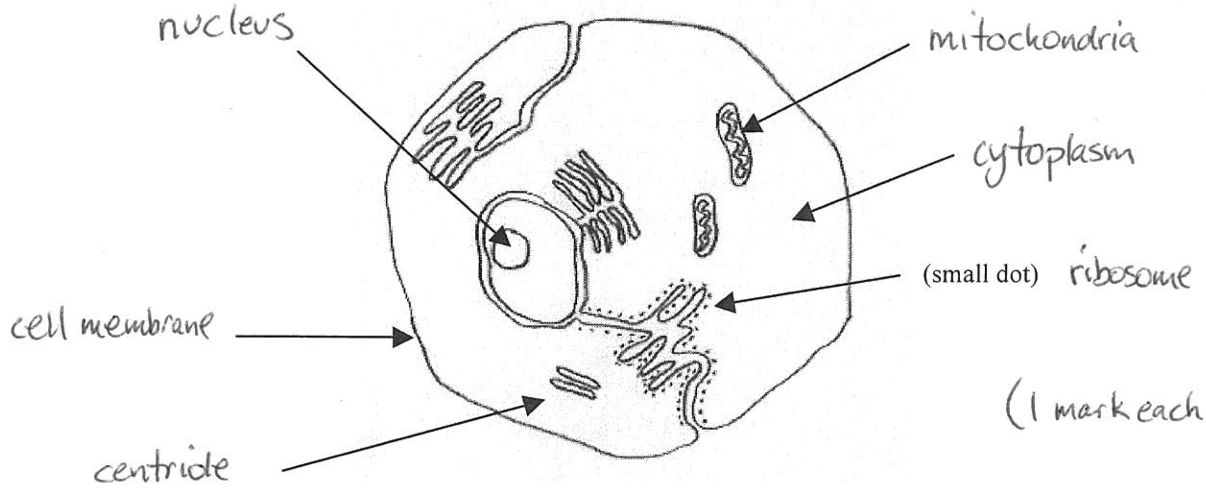
YEAR 10 SCIENCE
LIFE + LIVING
TEST - CELLS

NAME: SOLUTIONS MARK: —/35

Mark	ND	NW	C	O
Mark Range	<10	10-15	16 - 24	25 - 35

1. Label the parts of the cell as shown in the diagram below.



(1 mark each).

(6)

2. Complete the following table, giving the name of the part of the cell (animal or plant) responsible for the process.

Role Or Function In The Cell	Name of Part
Protects the cell and lets substances in and out.	cell membrane
Produce spindles for chromosome separation in cell division.	centriole
Produces energy for the cell.	mitochondrion
Controls the functions within the cell.	nucleus
Produces proteins for the cell.	ribosomes .

(5)

3. (a) Name the four bases in a DNA strand.

guanine

thymine

adenine

cytosine

(4)

(b) Match the 4 bases into 2 pairs of bases that go together. G - C

A - T (1 mark each)

(2)

(c) Draw a simple diagram to show how a short section of a chromosome would look. (***Draw only 5 pairs of bases. Choose any combination you like.)***

−A−T−G−C−A−

−T−A−C−G−T−

(2)

4. ***Mitosis*** occurs in normal body cells. Answer the following questions about this process.

(a) How many chromosomes are in a ***normal human cell***? 46

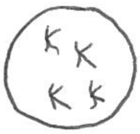
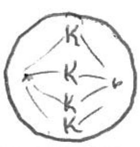
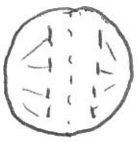
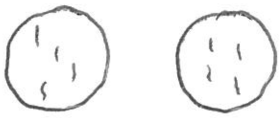
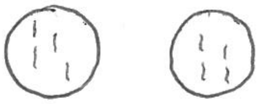
(b) At the end of mitosis, how many daughter cells are produced from one parent cell? 2

(c) How many chromosomes are found in each daughter cell? 46

(d) Is the answer to part (c) the ***haploid*** or ***diploid number***? diploid (1 mark each)

(4)

5. In the space below, draw the various stages in ***mitosis***. Next to each drawing, give ***the name of the stage*** and ***what is happening*** in the cell.

Stage Name	Drawing	Description
prophase		• Chromosomes become visible • They replicate (copy)
metaphase		• Line up across centre of cell. • Attached to spindle fibres.
anaphase		• Chromosomes are pulled apart.
telophase		• Cell pinches off into 2 parts.
interphase		• Rest phase

(5)

1 mark 2 marks 2 marks

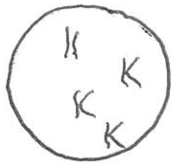
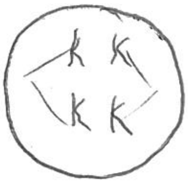
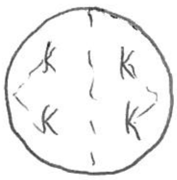
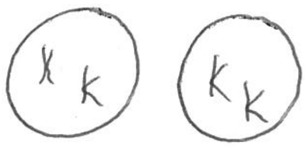
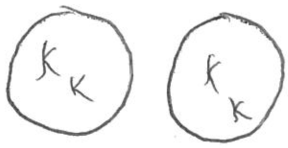
6. ***OPTIONAL***

Meiosis is an important process in the human body. It occurs in two parts.

(a) In which cells of the body does it occur? reproductive cells. (1)

(b) How many daughter cells does the process produce? 4 (1)

(c) Draw a series of simple diagrams to ***SHOW THE FIRST PART OF MEIOSIS***. Name each stage and state what is happening in the cell at each stage.

Stage Name	Drawing	Description
prophase I		• Chromosomes become visible. • Replicate (copy) themselves.
metaphase I		• Pairs of chromosomes line up across the centre.
anaphase I		• Pairs get separated by the spindle fibres.
telophase I		• Cell pinches off into 2 parts. • Haploid number of chromosomes.
interphase		• Rest period.

↑ 1 mark ↑ 2 marks ↑ 2 marks. (5)